

Project Title:

The Edge-Acoustic Launch Monitor: On-Device Audio
Classification for Golf Strike Quality

Course: EE 446: TinyML (Spring 2026)

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Problem Statement

Amateur golfers rely heavily on the acoustic feedback of a golf shot to evaluate strike quality, but quantifying that sound objectively requires expensive, highly complex acoustic launch monitors. A high-quality iron shot produces a distinct, sharp frequency, whereas a "chunk" (hitting the turf first) produces a low-frequency thud, and a "shank" (striking the hosel) creates a metallic ring. Sometimes golfers hit the ball with the wrong side of the club yet the ball flies, this makes it hard to know if the swing was good or not. Usually, they listen to the thud of the strike to know the type of shot taken.

This problem is highly relevant for TinyML because real-time acoustic analysis traditionally requires streaming raw audio to a cloud server or a tethered laptop, introducing latency and privacy concerns. By leveraging TinyML, we can perform on-device audio classification directly at the edge. A highly constrained, battery-powered microcontroller can be placed inches away from the tee on a simulator mat, processing the acoustic data locally to provide instant, zero-latency feedback on a player's mechanics without the need for external network infrastructure.

Dataset(s)

The project will utilize a custom, manually collected dataset generated using the built-in microphone on the Arduino Nano BLE. Data collection will span three golf simulator sessions to ensure natural variation in strike speed and ambient room conditions.

- **Classes:** The dataset will consist of three distinct categories: Pure_Strike, Chunk_Fat, Shank
- **Volume:** Approximately 150 total audio samples will be recorded (50 samples per class).
- **Key Features:** The raw audio will be sliced into 1-second time windows centered around the impact amplitude spike. To optimize for edge machine learning, the raw waveform will not be used directly. Instead, the audio will be converted into Mel-frequency cepstral coefficients (MFCCs). This DSP technique translates the time-series audio into a 2D spectrogram (a visual heatmap of the frequencies), highlighting the visual differences between the low-frequency thud of a chunked shot and the high-frequency ring of a shank.

Modeling Approach

- **Input Modality:** Audio (processed into 2D MFCC spectrograms).
- **Neural Network:** A standard 2D Convolutional Neural Network (CNN). The CNN is highly effective for this task because it will treat the MFCC spectrograms as an

image classification problem, scanning the 2D footprint for the visual shapes of the specific strike frequencies.

- **Compression Strategy:** Post-training 8-bit Integer Quantization (INT8) will be strictly enforced. The weights and activations of the CNN must be compressed from 32-bit floating-point numbers to 8-bit integers to drastically reduce the model's memory footprint, ensuring both the DSP pipeline and the model fit within the Arduino Nano BLE's 256KB of RAM.

Deployment Plan

The quantized model will be compiled into a C++ library and flashed permanently onto the Arduino Nano BLE. The hardware will be powered by a small 3.7V LiPo battery, acting as a completely untethered, standalone device.

For real-world evaluation, the Arduino will be placed approximately 12 inches away from the tee inside a golf simulator bay. The continuous microphone loop will trigger inference upon detecting a high-amplitude noise (the impact). The classification output will be evaluated visually using the board's built-in RGB LED. The LED will be programmed to flash a specific color based on the model's prediction (e.g., Green = Pure Strike, Red = Shank, Yellow = Chunk). The model's accuracy will be validated live by cross-referencing the LED output with the physical feel of the shot and the ball-flight data displayed on the simulator screen.

Challenges & Solutions

Challenge 1: Ambient Noise Interference Simulator bays contain significant background noise, including cooling fans, music, and adjacent players talking.

- **Solution:** The continuous audio buffer will utilize a strict volume threshold. The 1-second recording window will only trigger and save when the microphone detects the extreme decibel spike of a golf club impacting a ball or mat, effectively filtering out standard room noise.

Challenge 2: Hardware Memory Constraints Extracting MFCC features and running a neural network simultaneously can easily exhaust the 256KB SRAM on the Nano BLE, leading to board crashes.

- **Solution:** The audio sampling rate will be aggressively downsampled to 16 kHz, which is sufficient to capture the necessary frequencies of a golf strike while keeping the data buffers small. The CNN architecture will be kept shallow (fewer hidden layers), and INT8 quantization will be rigorously tested to guarantee execution within memory limits.

Challenge 3: False Positives from Practice Swings A player brushing the turf during a practice swing could accidentally trigger a "Chunk" classification.

- **Solution:** The dataset collection will prioritize the aggressive, high-speed impact of real strikes. If practice swings trigger false positives during early deployment testing, a fourth Practice swing/ Background class will be added to the dataset to train the model to explicitly ignore low-impact turf interactions.